Mechanism of the Structural Backfire Effect: How Hub Opinion Shifts Fragment Heavy-Tailed Social Networks

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Abstract

Online discourse on social media has become integral to public opinion formation for much of the population, and influencers (or hub users) exert a strong influence on online communication. We study a counterintuitive *structural backfire effect* in agent-based opinion dynamics: when a moderately opinionated hub is driven toward an extreme stance, it often fails to steer public posting dynamics and instead accelerates collective polarization by energizing opposite-opinion followers. We show that this is a robust structural phenomenon—emerging specifically in heavy-tailed networks when users are sufficiently stubborn and the opinion-shifting hub is sufficiently central, and absent in networks lacking a heavy tail. By directly measuring the causal chain, we identify its mechanism: hub manipulation severs the primary cross-partisan bridge, driving opposite-opinion followers to defect and collapse into echo chambers. The pre-manipulation model reproduces the empirical stylized facts of real social media data, and the effect persists under a realistic gradual opinion shift. These findings identify a candidate structural mechanism by which nonlinear network dynamics can turn a well-intentioned or adversarial opinion-shifting intervention into a self-defeating one in heavy-tailed networks with sufficiently stubborn users. Because these structural features are commonly observed in real social media platforms, our results suggest a possible source of vulnerability worthy of future empirical investigation.

Keywords: Opinion Dynamics, Social Media, Multi-Agent Simulation, Polarization, Echo Chamber

1 Introduction

The advent of social media platforms such as Facebook, X (formerly Twitter), YouTube, and TikTok has profoundly transformed the global information environment. By significantly reducing communication barriers, these platforms offer novel interaction modes in which finding like-minded communities is effortless and the cost of broadcasting to the public has drastically decreased [1–3]. Owing to their accessibility and efficiency, online social networks are increasingly envisioned as essential infrastructure for civic deliberation and digital democracy [4–6]. However, to fully harness the potential of these transformative technologies, we must understand the complex interplay between system features and user behavior, particularly because the relatively short history of human interaction with social media means that our understanding of its long-term societal impact remains nascent.

Despite these advantages, society faces significant challenges, including opinion polarization, misinformation diffusion, and malicious influence operations [7–9]. Although phenomena such as polarization are not exclusive to the digital age, the architectural and functional characteristics of social media arguably catalyze their emergence. Structural phenomena, such as *echo chambers* [10, 11], in which users are isolated in homophilic clusters driven by their own selective tie formation, and *filter bubbles* [12, 13], driven by algorithmic filtering, create environments that foster radicalization.

A less studied yet equally consequential source of polarization is the opinion shift of highly connected *hub users (or influencers)*, whose stance changes can propagate rapidly through the network. Such shifts arise spontaneously in events such as the sudden political revelations of prominent influencers, the ideological radicalization of public figures, or the adoption of increasingly extreme positions by populist politicians. Such events routinely trigger follower defection and affective polarization [14, 15]. Exogenous *influence operations* [7, 16, 17], which deliberately target opinion leaders to steer collective discourse, represent an intentional variant of the same mechanism.

Whether endogenous or exogenous, the structural and behavioral conditions under which such hub-driven opinion shifts trigger polarization remain imperfectly understood. While prior research has repeatedly shown that attempts to shift opinions through social influence frequently produce counterintuitive and nonlinear outcomes, including the reinforcement of prior positions rather than convergence toward the intended target [15, 18], how the network-structural conditions modulate these hub-mediated dynamics has not been sufficiently explained. In preliminary work, we proposed an *agent-based model* (ABM), which integrates well-established social interaction models with behavioral repulsion (tie severing through unfollowing rather than attitudinal reactance), and reported a novel network-level phenomenon termed the *structural backfire effect*, in which the opinion shift of moderately opinionated hub users toward extremes severs the cross-partisan bridge they provide, driving far-opinion followers to defect and retreat into echo chambers, ultimately intensifying polarization [19]. We use *structural* to distinguish this network-mediated effect from the classical, individual-level "backfire effect," in which exposure to opposing views reinforces a person's prior *opinion* [18, 20]: here, the intervention instead amplifies the opposing camp's *posting activity* through structural fragmentation of the network,

with little contribution from opinion-level repulsion term in the model. Hereafter we refer to it as the structural backfire effect, or simply backfire where unambiguous. However, these preliminary findings were limited to a specific parameter set, a single network topology, and two hub nodes undergoing instantaneous opinion shifts; moreover, the model lacked empirical grounding to establish its behavioral credibility relative to real social media dynamics. It therefore remained unclear whether the backfire effect is a universal structural phenomenon or an artifact arising only under narrow conditions.

This paper establishes the structural backfire effect as a robust phenomenon and explains its mechanism through four main contributions. First, we provide direct mechanistic evidence for the causal chain underlying the backfire effect by tracking unfollowing events, follower composition, and feed homogeneity at each manipulation step. This reveals echo chamber deepening via hub loss as a structural mechanism for backfire, which is distinct from the psychological reactance processes identified in prior literature [15]. Second, we identify the boundary conditions under which hub-driven opinion shifts trigger polarization through a systematic two-parameter sweep across agent stubbornness and opinion prevalence thresholds evaluated in heavy-tailed, high-clustering, and random networks. Third, we validate the model against empirical stylized facts drawn from Twitter/X data, establishing its behavioral credibility relative to real social media dynamics prior to the main analysis. Fourth, in contrast to the instantaneous opinion shift modeled in our preliminary study, we adopt a gradual shift model in which the hub's opinion is incrementally displaced per platform access, demonstrating that the backfire effect persists under this more realistic behavioral assumption. Together, these contributions establish the backfire effect as a robust structural phenomenon consistently associated with heavy-tailed degree distributions and high user stubbornness, providing a theoretical framework that highlights the critical role of hub users in network resilience and echo chamber formation.

2 Related Work

Computational social system modeling using ABM has long been a primary method for gaining insight into complex social phenomena. Early research modeled individuals as nodes in a graph with internal state variables and analyzed system convergence and dynamics through local interactions [21–25]. The *voter model* is the representative of these early researches in which agents hold binary values and probabilistically change their internal values by adopting the state of a neighbor. This model serves as a foundational framework to understand public opinion dynamics as a diffusion process on social networks.

These discrete models were extended by incorporating greater sociological realism [26–29]. DeGroot [26] introduced a weighted averaging model, in which an agent's opinion is updated based on the weighted average of their neighbors' opinions, representing the social influence process. Building on this concept, Deffuant et al. [28] and Hegselmann and Krause [29] introduced the concept of a *bounded confidence model*, which implies that agents interact only with those whose opinions fall within a specified

threshold range. This mechanism accounts for the psychological reality that individuals are unlikely to be influenced by divergent views, thereby providing a mathematical basis for the emergence of a local consensus or cluster.

Social media has introduced novel forms of communication environments that are distinct from traditional face-to-face interactions, characterized by features such as algorithmic recommender systems and dynamic tie formation (i.e., following and unfollowing relationships). The complex dynamics that emerge from these features pose critical questions that must be fully addressed; therefore, a vast body of literature has emerged to investigate these phenomena [9, 11, 30–36]. Vosoughi et al. [36] conducted a large-scale analysis of information diffusion on Twitter, and found that falsehoods spread more broadly, rapidly, and deeply than truths. They attribute this pattern primarily to the greater novelty of false information, which increases users' propensity to share such information. Furthermore, evidence indicates that humans, not automated accounts, serve as principal drivers of falsehood diffusion. Xie et al. [37] showed that a certain fraction of committed agents, who never change their opinions, can influence the overall opinion formation process by persuading individuals holding opposing views. Sasahara et al. [11] demonstrated that the interplay between social influence mechanisms and the unfollowing of social ties accelerates polarization, thereby facilitating the formation of echo chambers with the validation of their simulations against empirical data from X (formerly Twitter) regarding political polarization during elections [9], thus confirming qualitative alignment between the model and real-world data. More broadly, the information era has enabled researchers to leverage large-scale behavioral logs for comprehensive analysis and model validation.

In social network analysis, extensive research has focused on the role of influencers and hub nodes in social networks. A representative domain is *influence maximization*, which addresses the problem of selecting seed nodes to maximize the spread of information (i.e., "word-of-mouth") under resource constraints, primarily in the viral marketing context [38–41]. Kempe et al. [38] provided the first efficient algorithms with provable approximation guarantees for the influence maximization problem, which is defined as the NP-hard task of identifying a set of nodes to trigger the largest adoption cascade. More recently in tandem with the development of neural networks, Kumar et al. [41] employed graph embeddings and *graph neural networks* to efficiently solve this problem, demonstrating that deep-learning approaches outperformed classical heuristics.

Although these studies treated hub nodes as static conduits for information spread, they did not address the consequences of hub users' *own* opinion shifts, whether endogenous, as discussed in Section 1, or exogenously orchestrated through *influence operations* [42–45] such as bot-driven campaigns, strategic promotion, account hijacking, and cognitive warfare. Even within the influence operations literature, the emphasis has been on the attacking actors rather than on the network position of the targeted hubs: Ross et al. [43], for instance, found that the number and sophistication of bots outweigh their network position as drivers of opinion influence. As a result, few studies have systematically examined the network-structural conditions under which the opinion shift of a *human* hub user triggers polarization.

Table 1: Agent parameters and their value ranges

Parameter	Symbol	Range	Initial value
Opinion	$O_i(t)$	$[-1.0, 1.0]$	$\mathcal{N}(0.0, 0.6^2)$
Posting probability	$\rho_i(t)$	$[0.05, 0.5]$	0.1
Bounded confidence threshold	$\epsilon_i(t)$	$[0.2, 1.0]$	1.0

Our preliminary work identified a "backfire effect," in which the opinion shift of moderately opinionated hub users toward extremes, modeled as manipulation to isolate the structural mechanism, triggers a repulsion-driven backlash across the network [19]. However, these preliminary findings did not systematically examine the mechanism, boundary conditions, or robustness of this phenomenon— specifically, how the effect emerges structurally and how its emergence depends on the interplay between social modeling parameters and network topology. The present study addresses this gap by systematically varying agent stubbornness, opinion prevalence thresholds, and network structure across heavy-tailed, high-clustering, and random topologies.

3 Computational Social Simulation Model

3.1 Agent-Based Model (ABM)

We introduce an ABM designed to simulate the opinion dynamics on social media platforms. Let $G = (\mathcal{A}, E)$ be a directed graph representing the social network structure, where $\mathcal{A} = \{1, 2, \ldots, N\}$ denotes the set of N agents (i.e., social media users), and $E \subseteq \mathcal{A} \times \mathcal{A}$ denotes the set of directed edges representing follow relationships. A directed edge $(i, j) \in E$ indicates that agent i follows agent j. We define the adjacency matrix A such that $A_{ij} = 1$ if $(i, j) \in E$, and $A_{ij} = 0$ otherwise, where A_{ij} is the (i, j) element in the $N \times N$ matrix A. The dynamics simulations are executed in discrete time steps, denoted by $t \in \{0, 1, 2, \ldots\}$.

Agent $i \in \mathcal{A}$ is characterized by the internal parameters listed in Table 1. Its opinion $O_i(t) \in [-1, 1]$ represents the normalized stance of i on a specific issue at time t. Although opinions are inherently multidimensional, many studies in computational social science focus on a single salient issue and model opinions along a one-dimensional spectrum [11, 12]. Under this framework, $O_i(t)$ represents agent i's stance toward a specific controversial topic, such as support for or opposition to a particular *hashtag* on social media. Similar issue-specific representations are commonly employed in empirical analyses of online political communication and polarization [9, 46].

Boundaries -1 and 1 correspond to the most extreme opposing views, and 0 represents a neutral or moderate attitude. We denote the initial opinion $O_i(0)$ as the agent's *intrinsic opinion*, representing its baseline belief that may influence its dynamics over time [27], and set it to a random number drawn from a Gaussian distribution with mean $\mu = 0.0$ and standard deviation $\sigma = 0.6$, clipped to the valid range $[-1, 1]$. To ensure a well-defined polarized environment, the two highest-degree agents are initialized with intrinsic opinions of $+1.0$ and -1.0. This setting approximates highly

polarized online discussions in which influential users occupy opposing communities. The manipulated hub is selected only after the network reaches a steady state and is never one of the seeded agents, so the initialization establishes the polarized context without specifying the manipulation target.

The posting probability $\rho_i(t)$ defines the likelihood that agent i posts at time t. This parameter corresponds to the concept of *intensity*, which is often used in social simulations. In our model, posting intensity ($\rho_i(t)$) is defined independently of opinion extremeness ($|O_i(t)|$). That is, activity is not proportional to radicalization. This independence allows us to capture heterogeneous behavioral patterns, including the phenomenon in which moderate users intensify their posting activity.

As described in Section 2, *bounded confidence* captures the sociopsychological hypothesis that individuals are unlikely to engage with opinions that deviate significantly from their own. Unlike classical bounded confidence models, in which the threshold directly gates the opinion-averaging step [28, 29], in our model the opinion update (Eq. (1)) averages over all posts currently in the feed. This reflects the assumption that a user is inevitably influenced by any information they actually read: once a post enters the feed and is read, it affects the reader's opinion regardless of how far it lies from their own. Bounded confidence therefore does not filter the opinion update itself; instead it governs the tie-level behaviors (reposting, following, unfollowing, and confidence decline) and thereby shapes the opinion dynamics indirectly, by determining which posts enter the feed in the first place. We further introduce a *dynamic* bounded confidence parameter $\epsilon_i(t)$ rather than a static threshold, modeling the adaptive evolution of user openness to conflicting views through interactions on the platform. The evolution of these parameters is governed by the social update mechanisms, as described next.

3.2 Modeling Social Adaptation

In general, humans tend to adapt their internal state to their environment. We model these mechanisms based on well-established computational theories [11, 27–29]. Each time an agent accesses the platform with probability P_a, it updates the internal parameters according to the rules described below. Hereafter, we call a post p_j from agent j *comfortable* for agent i when $|O_i(t) - O(p_j)| \leq \epsilon_{\min}$, where $\epsilon_{\min} = 0.2$ is the minimum bounded confidence threshold (i.e., the lower bound of $\epsilon_i(t)$); this notion is used by the social-reinforcement rule below. The dynamic threshold $\epsilon_i(t)$ itself instead governs the tie-level behaviors (reposting, following, unfollowing, and confidence decline), as specified in each rule.

Social Influence. Agents read all posts in their feeds and update their opinions based on posts in feeds and their intrinsic opinions. Agent i updates its opinion at time t, influenced by the opinion of post p_j in its feed $F_i(t)$, as follows:

$$O_i(t) = \mu_{stub} \cdot O_i(0) + (1 - \mu_{stub}) \cdot \frac{\sum_{p_j \in F_i(t)} O(p_j)}{|F_i(t)|}, \tag{1}$$

where μ_{stub} denotes the agent's *stubbornness*, representing the extent to which agents adhere to their intrinsic opinions $O_i(0)$ rather than being influenced by others [27]. Note that posts in an agent's feed originate from those posted or reposted by users whom the agent follows; thus, the agent may encounter posts that lie beyond its current bounded confidence unless it unfollows or blocks the creator.

Social Reinforcement. Agents increase or decrease their posting probability (or intensity) based on the ratio of comfortable posts in their feeds. Formally, let $C_i(t) = |K_i(t)|/|F_i(t)|$ be the ratio of comfortable posts in agent i's feed at time t, where $K_i(t) = \{p_j \in F_i(t) \mid |O_i(t) - O(p_j)| \leq \epsilon_{\min}\}$ denotes the set of comfortable posts, as defined above. The posting probability is then updated by

$$\rho_i(t) = \begin{cases} \min(\rho_i(t-1) \cdot (1+\delta_\rho),\ 0.5) & \text{if } C_i(t) > \varepsilon_s, \\ \max(\rho_i(t-1) \cdot (1-\delta_\rho),\ 0.05) & \text{else if } C_i(t) \leq 1-\varepsilon_s, \\ \rho_i(t-1) & \text{otherwise,} \end{cases} \tag{2}$$

where δ_ρ (> 0) denotes the parameter controlling the posting intensity adjustment rate, and $\varepsilon_s \in (0,1)$ denotes the *threshold for opinion prevalence*. An agent intensifies its posting when comfortable content locally prevails ($C_i(t) > \varepsilon_s$) and reduces it when comfortable content is locally scarce ($C_i(t) \leq 1 - \varepsilon_s$). The two conditions are evaluated in order, so the increase rule takes precedence; consequently, when $\varepsilon_s > 0.5$, an intermediate band $1 - \varepsilon_s < C_i(t) \leq \varepsilon_s$ leaves the posting probability unchanged.

Social Reward. In addition to the conformity-based reinforcement above, agents receive a posting incentive proportional to the social acceptance of their posts. In our implementation, this reward is applied at the *start* of each platform access, before the agent reads its current feed, and reflects the reposts accumulated since the previous access. When agent i's posts are reposted by others, i updates its posting probability by a reward r_i based on the principle of diminishing marginal utility [47]:

$$r_i = \alpha \cdot \ln(1 + N_i^l), \tag{3}$$

where $\alpha > 0$ is a tuning parameter and N_i^l is the number of reposts received since i's previous platform access. The posting probability is then updated as $\rho_i(t) \leftarrow \min(\rho_i(t) \cdot (1 + r_i),\ 0.5)$. This logarithmic formulation reflects the empirically observed saturation of social validation effects at high repost counts.

Confidence Decline. The update of the bounded confidence threshold reflects the asymmetric human tendency for negative experiences to carry greater psychological weight than positive ones [48, 49]; accordingly, exposure to conflicting content erodes openness whereas comfortable content does not symmetrically restore it, causing ϵ_i to evolve monotonically downward. Concretely, agents decrease their bounded confidence for *each* post they read that lies beyond their current bounded confidence in a given step. Formally, ϵ_i is decremented multiplicatively by δ_ϵ for each post $p_j \in F_i(t)$ that satisfies $|O_i(t) - O(p_j)| > \epsilon_i$, where ϵ_i takes the value already updated by preceding posts within the same step:

$$\epsilon_i \leftarrow \max(\epsilon_i \cdot \delta_\epsilon,\ 0.2) \quad \text{for each } p_j \in F_i(t) \text{ s.t. } |O_i(t) - O(p_j)| > \epsilon_i. \tag{4}$$

Thus, a single step may produce multiple successive decays if the feed contains multiple such posts. The parameter δ_ϵ ($0 < \delta_\epsilon < 1$) denotes the multiplicative decay rate of confidence in the surrounding environment.

3.3 Modeling Behaviors on Social Media

As this study specifically focused on the dynamics within social media, we introduce the following characteristic behaviors: post, repost (like), follow, and unfollow (block).

Post. Every time agent $j \in \mathcal{A}$ accesses the platform, it creates a post p_j reflecting its current opinion with probability $\rho_j(t)$. If j creates a post, it is added to the feeds of agents who follow j; each agent reads at most S_{fd} posts per platform access, in reverse-chronological (newest-first) order, reflecting cognitive limits on information intake.

Repost (Like). When agent $i \in \mathcal{A}$ reads its feed, it reposts each post that lies within its bounded confidence (i.e., $|O_i - O_{p_j}| < \epsilon_i$) independently with probability P_r. Posts are distributed across the social media network via this repost function; posts reposted by an agent are added to the feeds of its followers. Note that the "Like" function on real-world social media services has a distribution effect similar to that described here; therefore, we do not distinguish between Like and Repost in the model for simplicity.

Follow. After reading the feed, agent i collects the creators of all posts within its bounded confidence (i.e., $|O_i - O_{p_j}| < \epsilon_i$) with no existing follow connection as candidates; with probability P_f, i randomly selects and follows one candidate. The maximum number of agents that any agent can follow is fixed at S_{fol} throughout the experiments; in this study, we set $S_{fol} = 10$ based on the concept of limited attention. Although users may follow hundreds of other users, the number of information sources that can effectively influence them at any given time is constrained by human cognitive limits and information overload [50, 51]. Accordingly, when an agent attempts to follow a new agent while already at S_{fol}, it unfollows the single currently-followed agent whose opinion is farthest from its own.

Unfollow (Block). Conversely, if a post lies beyond an agent's bounded confidence (i.e., $|O_i - O_{p_j}| > \epsilon_i$), the agent unfollows the creator with probability P_u, and rejects all future posts. If no connection exists, the agent blocks the creator with probability P_u, and rejects all future posts from the creator.

While all agents share the same behavioral rules, heterogeneity arises endogenously through individually evolving state variables, distinct initial opinions, and the agents' positions; for instance, hub nodes and peripheral nodes experience qualitatively different interaction dynamics even under identical rules.

3.4 Modeling Manipulation

The primary objective of this study is to understand the nonlinear dynamics in which an influential *hub* user starts to shift their opinion toward an extreme position. Accordingly, we manipulated a single hub user's opinion toward a designated extreme in our simulations. The manipulation target was selected based on ideological position and

structural importance in the network. After the social media network had evolved sufficiently to a steady state, agents with neutral opinions (i.e., $|O_i| < 0.2$) and follower counts of at least 10% of the total population (i.e., ≥ 100 agents for $N = 1000$) were identified as candidates. One agent was then randomly selected from this pool as the manipulation target. If no qualifying candidate was identified, the simulation run was excluded from the results.

We modeled this manipulation as a gradual opinion shift: each time the target agent accesses the platform, its opinion is incremented by a fixed displacement $\Delta = 0.01$ in the target direction, bypassing the social influence update in Eq. (1), until it reaches an extreme value ($+1.0$ or -1.0). The target's posting probability is simultaneously fixed at the maximum value $\rho_{\max} = 0.5$ to reflect the heightened activity of a manipulated influencer. Under a platform access probability of $P_a = 0.1$, the expected number of time steps for the target to reach the extreme is approximately 1,000 steps, given that the target's opinion is in $[-0.2, 0.2]$. In contrast to the instantaneous shift modeled in our preliminary study [19], this gradual formulation captures a more realistic process of opinion radicalization. Although highly simplified, this single-agent setting abstracts away the operational mechanism by which a hub's opinion shifts, whether through spontaneous radicalization or coordinated external manipulation, and thereby isolates the structural mechanism driving the backfire effect.

4 Experiments and Results

4.1 Experimental Settings

To evaluate the robustness and identify the boundary conditions of the backfire effect, we conducted a sensitivity analysis across five network configurations generated by four distinct models: the *connecting nearest neighbor with random linkage* (CNNR) [52], the *Holme–Kim* (HK) [53], the *Watts–Strogatz* (WS) [54], and the *Erdős–Rényi* (ER) [55] models.

The CNNR model extends the original CNN topology [56] by incorporating randomness to simulate human relationship formation on social media platforms, reproducing preferential attachment, hierarchical clustering, and heavy-tailed degree distributions. For the HK network, two conditions were simulated, namely $p_t = 0$ and 0.3. The HK model combines preferential attachment with a triangle-closing mechanism controlled by the parameter p_t: setting $p_t = 0$ recovers pure preferential attachment (yielding a network equivalent to the Barabási–Albert model [57] with low clustering), while $p_t = 0.3$ additionally closes each newly added edge into a triangle with probability p_t, generating a heavy-tailed network with substantially higher clustering. These two HK variants thus allow us to disentangle the contributions of a heavy-tailed (hub-dominated) degree structure and local clustering within the same generative framework. The WS model constructs a small-world network by rewiring a regular ring lattice with rewiring probability β, resulting in a topology that exhibits high clustering and short average path lengths but lacks a heavy-tailed degree distribution. The ER model serves as a random baseline lacking both hub structure and clustering. The generation parameters and their resulting average network metrics are summarized in Table 2.

Table 2: Network metrics of the initial networks (mean over 20 seeds; path length, diameter, and the power-law exponent over 5 seeds).

Network topology	CNNR	HK (p_t=0.0)	HK (p_t=0.3)	WS	ER
Number of agents	1000	1000	1000	1000	1000
Average degree	4.0	4.0	4.0	4.0	4.0
Average path length	5.06	3.58	3.26	8.78	5.11
Diameter	14	7	7	17	10
Power-law exponent	2.26	2.13	2.03	—	—
Average clustering coefficient	0.22	0.05	0.21	0.18	0.002

Crucially, to ensure that any difference in the backfire effect across these topologies reflects network *structure* rather than a confound of overall connectivity, we calibrated each model's generation parameter so that all five networks share a common undirected mean degree $\langle k \rangle = 2E/N \approx 4$ (Table 2). The topologies therefore differ in their degree *distribution*, clustering, and path structure while holding edge density fixed, so that mean degree cannot account for the topology contrast reported below. As Table 2 shows, under this alignment CNNR and both HK variants retain heavy-tailed in-degree distributions (power-law exponent $\approx$ 2.0–2.3, estimated by maximum likelihood with the `powerlaw` package), whereas WS and ER do not. This contrast persists through the pre-manipulation evolution: the fitted exponent at the manipulation onset (step 20,000) remains low for CNNR/HK ($\approx$ 1.8–1.9) but rises sharply for WS/ER ($\approx$ 3.8–4.1), confirming that no heavy tail emerges endogenously in the latter two.

The parameters and their values are listed in Table 3. The intensity adjustment rate δ_ρ and confidence decay rate δ_ϵ were set based on qualitative reasoning about known social dynamics: bounded confidence erodes gradually under sustained exposure to opposing views, and posting intensity adjusts incrementally in response to social feedback. The empirical adequacy of the resulting model is assessed independently in Section 4.2. Each simulation run involved 1,000 agents over 40,000 steps, with the manipulation phase introduced at step 20,000. The code is publicly available on GitHub.[1]

4.2 Empirical Validation via Stylized Facts

Agent-based models of social phenomena cannot typically be validated through direct predictive testing against held-out data; the established alternative is to verify that the model reproduces *stylized facts*, which are qualitative regularities robustly documented in the empirical literature, rather than fitting specific numerical targets [58]. We adopt this approach and assess the pre-manipulation equilibrium of our model against three stylized facts drawn from empirical studies of Twitter/X, using $n = 30$ complete simulation runs under the baseline parameter settings ($\varepsilon_s = 0.5$, $\mu_{stub} = 0.7$, and the Holme–Kim network with $p_t = 0.3$).

[1] https://github.com/tooooomoya/structural-backfire-effect

Table 3: Experimental parameters

Parameter	Symbol	Value
Intensity Adjustment Rate	δ_ρ	0.1
Social Reward Tuning Parameter	α	0.01
Confidence Decay Rate	δ_ϵ	0.99
Maximum Number of Followees	S_{fol}	10
Maximum Feed Size	S_{fd}	10
Threshold of Opinion Prevalence	ε_s	Variable
Agent's Stubbornness	μ_{stub}	Variable
Probability of Accessing the Platform	P_a	0.1
Probability of Reposting	P_r	0.25
Probability of Following	P_f	0.1
Probability of Unfollowing	P_u	0.1

SF1: Apparent polarization. The variance in observable post content across users consistently exceeds the variance in the underlying opinion distribution, because users with more extreme views post disproportionately more frequently [59]. We operationalize this phenomenon as the ratio $\sigma^2_{\text{post}}/\sigma^2_{\text{opinion}}$, where $\sigma^2_{\text{opinion}}$ is the opinion variance weighted by the number of users at each opinion position (the distribution of opinion holders) and σ^2_{post} is the same variance weighted by the number of posts contributed at each position (the distribution of observable content). To anchor this ratio empirically, we recompute it directly from the publicly released account-level data of [59] ($n = 762$ surveyed Twitter/X users with self-reported ideological leans on a $[-3, +3]$ scale, contributing $\sim$1.96M posts over 2020–2021).

In these data, hyper-partisan accounts post roughly two to three times as often as moderates, inflating the post-weighted opinion variance relative to the user-weighted variance by a factor of 1.19 over the full period and rising to 1.26 during the high-salience 2020 election window. This latter period represents precisely the kind of high-attention regime that our manipulation scenario is designed to capture. This yields an empirical target range of $[1.19, 1.26]$. Our model achieves a median ratio of 1.248 (IQR $[1.175, 1.299]$), with 100% of runs exceeding the threshold of 1.0. The median falls within the empirically derived range, consistent with the stylized fact that publicly observable opinion is more dispersed, and thus appears more polarized, than the underlying opinion distribution.

SF2: Cross-partisan repost fraction. Repost (retweet) networks are strongly segregated along partisan lines: cross-partisan reposting is near-zero, far below the rate expected under random mixing. Conover et al. [9] report a retweet-network modularity of 0.48 (vs. 0.17 for the mention network) and find that cross-ideological retweet links occur at only 3–5% of their randomly-expected frequency, concluding that ideologically opposed users "very rarely share information from across the divide." As an upper reference, Wojcieszak et al. [60] find that users share content from out-group political elites only about one-fourteenth as often as from in-group elites ($\approx$ 7% of partisan shares crossing the divide). We operationalize this as the fraction of repost edges crossing the opinion midpoint among partisan agents. Our model yields a median cross-partisan repost fraction of 6.1% (IQR $[3.8\%, 9.8\%]$), which sits in the near-zero

regime, falling below the $\approx 7\%$ upper reference, consistent with the strong segregation documented above. This supports the validity of the dynamic bounded confidence and unfollowing mechanisms in generating realistic levels of within-group information sharing.

SF3: Echo chamber depth. Politically engaged users inhabit *partial* echo chambers: a majority, but not the entirety, of their information diet is ideologically aligned. Cinelli et al. [10] show that, on Twitter, a user's leaning is strongly correlated with the average leaning of their network neighborhood, while Bakshy et al. [34] find that only about 29.5% (24% for liberals, 35% for conservatives) of the hard-news content in a user's feed is cross-cutting, meaning that roughly 70% is ideologically aligned, and the partisan retweet communities of Conover et al. [9] are 80–93% homogeneous. We operationalize echo chamber depth as the within-group comfort rate (the aligned fraction of the feed) for extreme-opinion agents. Our model yields a median of 52.5% (IQR $[44.7\%, 61.4\%]$), which sits just below the empirical aligned-share range for engaged users ($\approx$65–76%; [34]). We therefore read SF3 as a *directional*, not a point match: the model reproduces the qualitative signature of a partial echo chamber characterized by a majority-aligned but far-from-total information diet, while producing a slightly shallower chamber than observed. If anything, this milder homophily makes the model a conservative substrate for the backfire effect rather than one that inflates it.

The model quantitatively reproduces SF1 and SF2 and matches the qualitative direction of SF3 (a partial, slightly shallower echo chamber), thereby providing empirical credibility for the model's structural assumptions in the pre-manipulation equilibrium.

4.3 Appearance of the Structural Backfire Effect

Consistent with our previous study [19], we first confirmed the appearance of the backfire effect, here in a Holme–Kim network with $p_t = 0.3$, by setting $\varepsilon_s = 0.5$ and $\mu_{stub} = 0.7$. Figure 1 shows the time-series evolution of a per-group log-response. For each opinion camp g, we plot $\log\big[\, v_g(t) \,/\, \bar{v}_g^{\text{pre}} \,\big]$, which is the logarithm of its instantaneous posting volume $v_g(t)$ normalized by its own mean volume of posts over the pre-manipulation window $t \in [19{,}000, 20{,}000]$. This normalization removes the large baseline differences in posting volume between camps, so that the manipulation-induced change itself becomes the signal. The three camps are defined relative to the manipulation direction: the *target* camp (the extreme side toward which the hub is driven, $|O_i| > 0.2$ on the target sign), the *opposite* camp (the opposing extreme), and the *neutral* camp ($-0.2 \le O_i \le 0.2$). To ensure the robustness of the observed trends, the curves are averaged across all 43 qualifying seeds for this condition and pooled over both manipulation directions (± 1.0), resulting in 86 trials, with shaded bands denoting ± 1 standard error. The manipulation was initiated at the 20,000th step, thereafter gradually shifting the targeted hub's opinion toward its assigned extreme with the aim of influencing public opinion on that side. We summarize each trial by a scalar backfire metric: writing $\bar{v}_g^{\text{post}}$ for the mean posting volume of camp g over the post-manipulation window $t \in [39{,}000, 40{,}000]$ (and $\bar{v}_g^{\text{pre}}$ for the pre-manipulation window defined above), the per-trial backfire metric is

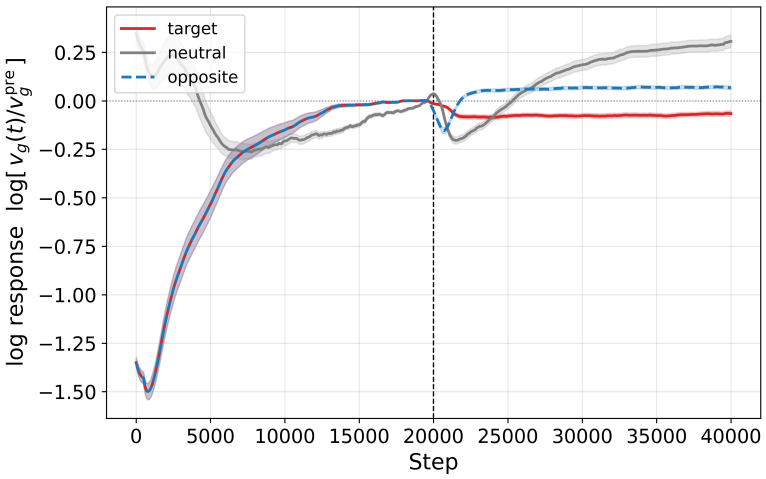


Fig. 1: Appearance of the backfire effect in a Holme–Kim network ($p_t = 0.3$, $\mu_{stub} = 0.7$, $\varepsilon_s = 0.5$). Curves show the per-camp log-response $\log[v_g(t)/\bar{v}_g^{\mathrm{pre}}]$, i.e., the posting volume of the target (solid red), neutral (solid gray), and opposite (dashed blue) camps, each normalized by that camp's mean volume over the pre-manipulation window $t \in [19{,}000, 20{,}000]$. The dashed vertical line marks the manipulation onset at step 20,000, and the dotted horizontal line marks the pre-manipulation baseline. Curves are averaged over 86 trials (43 seeds $\times$ two manipulation directions, all qualifying seeds for this condition), with shaded bands denoting ± 1 standard error.

$$b = \log\left(\frac{\bar{v}_{\mathrm{opp}}^{\mathrm{post}}}{\bar{v}_{\mathrm{opp}}^{\mathrm{pre}}}\right) - \log\left(\frac{\bar{v}_{\mathrm{tgt}}^{\mathrm{post}}}{\bar{v}_{\mathrm{tgt}}^{\mathrm{pre}}}\right), \tag{5}$$

a difference-in-differences estimator on the log scale, where a positive value of b indicates that the opposite camp grew more active than the target camp relative to the baseline, confirming the occurrence of the backfire effect. By construction, the gap between the opposite and target curves in Figure 1 is the time-resolved metric $b(t)$, which serves as the running analogue of the scalar b.

As can be seen, during the first half of the simulation, prior to any manipulation, the two conflicting opinionated camps (target and opposite, which are statistically symmetric at this stage) grow in posting activity by a substantially larger factor than the neutral camp, as evidenced by their steeper rise toward the baseline. This reflects the steady emergence of polarization: as echo chambers form on both extremes, agents on the extremes come to post increasingly often relative to the moderate majority. We note that, because each curve is normalized to its own pre-manipulation window, all three necessarily converge at the onset by construction; the informative content is therefore the relative steepness of the curves before this point, and, most importantly, their divergence after it.

In the latter half, although intervening on a neutral hub to drive it toward the target extreme would intuitively be expected to amplify only the target camp, the

opposite was observed. Immediately after the onset, the neutral camp's posting volume collapsed sharply, dropping to roughly 0.2 log units below its baseline, before recovering and eventually exceeding its pre-manipulation level. Furthermore, after a brief initial transient in which the target camp ticked up, the opposite camp's log-response rose above that of the target camp and remained elevated long after the manipulation finished at around the 21,000th step. Equivalently, the difference $b(t)$ became and stayed positive: the opposite camp grew more active, relative to its own pre-manipulation baseline, than the very camp the manipulation sought to promote. Because this pattern is averaged over both manipulation directions and all 43 qualifying seeds, it is a symmetric effect rather than a one-sided artifact of a particular target sign. This sustained, counterintuitive outcome, which runs contrary to the intended purpose of the intervention, is what we term the *structural backfire effect*.

4.4 Mechanism of the Structural Backfire Effect

Regarding the backfire effect confirmed in Section 4.3, we hypothesize a four-link causal chain: an opinion shift by the targeted moderately opinionated hub (1) drives its opposing followers to unfollow it, which (2) strips the network of the bridge that connected the opposing camp to diverse information, such that (3) the opposing camp collapses into a more pure echo chamber, and (4) this homogenization raises the opposing camp's posting activity until it outgrows the targeted camp, producing the backfire effect. Because this plausible narrative is not self-evident, this section measures each link directly on the canonical backfire baseline (Holme–Kim with p_t=0.3, μ_{stub}=0.7, ε_s=0.5; using n=86 trials from 43 seeds $\times$ two manipulation directions). The per-trial outcome is the backfire metric b of Eq. (5); over these trials the backfire effect is present and robust ($\bar{b} = +0.135$, seed-clustered Cohen's $d = +1.00$, one-sided t-test vs. 0: $p < 10^{-7}$). As shown later in the dose–response analysis (Section 4.5), this baseline is a conservative operating point: the effect grows further with the manipulated hub's in-degree. All quantities below are derived from simulation snapshots. To enable comparison across manipulation directions, opinion classes are defined relative to the manipulated side: the camp containing the manipulated hub is labeled "targeted," whereas the opposite camp is labeled "opposing." This convention allows results from both manipulation directions to be pooled.

4.4.1 Selective unfollowing

Unfollowing of the manipulation targeted hub is sharply asymmetric and concentrated after the intervention (Table 4). In the pre-manipulation window, the two camps unfollowed the hub at equal rates (13.9 unfollows each), whereas after the manipulation, the opposing camp unfollowed it nearly four times as frequently as the targeted camp (117.7 vs. 29.5 per trial; paired $t = 12.4$, $p < 10^{-16}$ for the opposing camp's pre-to-post increase). As a result, the hub's follower set, which was balanced at the onset (33% opposing, 33% targeted, 33% neutral), was purged of opposing followers by the end of the run (0% opposing, 97% targeted), thereby converting the manipulated hub from a cross-cutting broadcaster into a same-side echo node. Across the same trials, the opposing and targeted camps remained noticeably more balanced in

Table 4: Quantitative evidence for the backfire causal chain (HK p_t=0.3, μ_{stub}=0.7, ε_s=0.5). Values are mean ± standard error across trials, pooled over both manipulation directions.

Causal Chain	Quantity	pre/onset	post/end
(1)	Opposing-camp unfollows of hub	13.9 ± 1.3	117.7 ± 9.0
(1)	Targeted-camp unfollows of hub	13.9 ± 1.3	29.5 ± 2.1
(1)	Opposing frac. of hub followers	0.33 ± 0.01	0.00 ± 0.00
(1)	Targeted frac. of hub followers	0.33 ± 0.01	0.97 ± 0.00
(2)	Hub betweenness (giant component)	0.183 ± 0.018	0.033 ± 0.006
(2)	λ_2 (giant component)	0.366 ± 0.012	0.284 ± 0.010
(3)	Opposing-camp comfort post rate (aggregate)	0.472 ± 0.002	0.497 ± 0.002
(3)	Comfort post rate, hub's former followers (opp.)	0.48 ± 0.02	0.60 ± 0.02
(3)	Comfort post rate, non-followers (opp.)	0.56 ± 0.02	0.55 ± 0.02
(3)	Network modularity	0.43 ± 0.00	0.49 ± 0.01
(4)	Opposing-camp post share	0.453 ± 0.006	0.459 ± 0.013
(4)	Targeted-camp post share	0.452 ± 0.007	0.415 ± 0.013

size than the underlying unfollowing gap (opposing/targeted population ratio: 1.03 pre-manipulation vs. 1.18 post-manipulation), confirming that the unfollowing gap (a nearly four-fold difference in raw counts) far exceeds any compositional shift between camps (an 18% population-ratio change).

4.4.2 Loss of the bridging role

Initially, the hub serves as a structural bridge connecting users with heterogeneous political opinions. As the manipulation progresses, it gradually loses this bridging role due to the loss of cross-cutting follower ties. Consistent with this interpretation, the hub's betweenness centrality, measured on the undirected giant component, drops more than five-fold (0.183 to 0.033), providing strong structural evidence that it no longer functions as a bridge. At the global level, the algebraic connectivity (λ_2) of the giant component also decreases (0.366 to 0.284), indicating a reduction in overall network cohesion. Although echo chambers have already formed to some extent by the onset of the manipulation, this additional decline in λ_2 suggests that the loss of the hub's cross-cutting ties further fragments the remaining network.

4.4.3 Echo chamber formation

The decisive evidence for the mechanism is its *specificity*. At the camp level, the effect appears muted: the opposing camp's mean comfort post rate, which is the aligned fraction of the feed defined via $\epsilon_{\min}$ in the social reinforcement rule, rose only modestly (from 0.472 to 0.497). On the other hand, disaggregating the data by prior attachment to the hub reveals the underlying mechanism. Among opposing agents, those who *were following the hub* at the onset saw their comfort post rate rise from 0.48 to 0.60 once they unfollowed it, approaching the level of opposing agents who were *not* following the hub, whose rates remained flat at around 0.55. Structurally, the network's modularity

rose over the same interval (from 0.43 to 0.49), echoing this localized homogenization. In other words, the hub was precisely what kept its opposing followers exposed to cross-cutting content; severing these ties collapsed them into homogeneous feeds, while agents who never relied on the hub were mostly unaffected. This ties the echo chamber to the hub disconnection rather than to generic repulsion dynamics.

4.4.4 From echo chamber to posting activity

Because the backfire effect is defined in terms of posting volume, the chain must terminate in posting activity. The fraction of agents that increase their posting probability after the intervention is higher in the opposing and neutral camps (0.25 and 0.25, respectively) than in the targeted camp (0.21). Moreover, the realized post share of the opposing camp rises (from 0.453 to 0.459) while the targeted camp's share falls (from 0.452 to 0.415), an asymmetry that is captured by $b > 0$. Because the manipulation bundles the opinion shift with the fixed activity level $\rho_{\max} = 0.5$ (Section 3) as a single realistic treatment, it is worth noting which component drives the asymmetry above: both camps are equally exposed to the hub's heightened activity, yet only the opposing camp's unfollowing rises sharply (Section 4.4.1), pointing to opinion distance rather than exposure volume as the proximate driver within this bundled treatment.

4.4.5 Dose-response identification

Although the results above show the average effect over the trials, this still leaves open whether the proposed causal chain operates on a trial-by-trial basis. We test this directly by asking whether the trial-level backfire metric b scales with the trial-level intensity of two upstream links across the 86 trials of this baseline: (i) the number of opposing-camp unfollows of the hub over the post-manipulation window (steps 20,000–40,000; link 1), and (ii) the trial's rise in the opposing camp's aggregate comfort post rate between the pre- and post-manipulation windows (link 3). Both are positively and significantly correlated with b (Spearman $\rho = +0.65$, $p < 0.001$ for unfollowing; $\rho = +0.55$, $p < 0.001$ for the comfort-post-rate rise; both $n = 86$), indicating that trials with more pronounced disruption of the hub's follower base and a larger shift toward the opposing camp's echo chamber also exhibit a larger backfire effect. The same graded logic extends to the hub's own structural prominence: the per-trial backfire metric b also increases monotonically with the target's in-degree at the onset of the manipulation, as quantified across the heavy-tailed regime in Table 5 (Section 4.5). Taken together, these within- and across-condition dose-response results provide convergent evidence consistent with the proposed causal chain and strengthen the interpretation that this pathway contributes to the observed backfire effect within the present model.

4.5 Sensitivity Analysis

To assess the robustness of the backfire effect and delineate its boundary conditions, we conducted a sensitivity analysis across five network configurations generated by four structural models (CNNR, HK with p_t=0.0 and 0.3, WS, and ER) by varying two key parameters: the threshold of opinion prevalence (ε_s) and social influence

stubbornness (μ_{stub}). The magnitude of the backfire effect was quantified using the same per-trial backfire metric b defined in Eq. (5) (see Section 4.3), where a positive value of b indicates that the opposing camp grew more active in posting (relative to its pre-manipulation baseline) than the targeted camp—the signature of the backfire effect. Each parameter condition was evaluated over up to 100 simulation runs (50 independent seeds $\times$ two manipulation directions). Because a subset of runs lacked a qualifying manipulation target (Section 3), the realized per-condition sample size is $\min(n_{\text{valid}}, 50)$—equal to the full 50 seeds in 103 of the 125 parameter$\times$network conditions and falling as low as 37 in the remaining 22 (full breakdown in Appendix A)—which does not bias the conclusions below, since each cell's Cohen's d and significance are estimated from its own realized sample and never compared against another network or condition using a formal cross-condition statistic. Because the two manipulation directions for a given seed are executed on the same network realization, they do not constitute independent observations. We therefore first average the per-trial backfire metric b over the two directions within each seed, to obtain a single value per seed, and subsequently report the effect size as Cohen's d computed from these seed-level values.

Statistical significance was assessed via a one-sample t-test of these seed-level values against zero (one-sided, alternative: mean > 0). Rejection of the null hypothesis indicates that the opposing camp systematically outgrew the targeted camp following the manipulation, thereby confirming the backfire effect. We report uncorrected significance at the 5% ($p < 0.05$), 1% ($p < 0.01$), and 0.1% ($p < 0.001$) levels. Because each network's heatmap comprises 25 simultaneous comparisons, we additionally control the false discovery rate with a Benjamini–Hochberg correction within each 25-cell family; the asterisks in Figure 2 mark cells that remain significant after this correction ($^{*}\, q < 0.05$; $^{**}\, q < 0.01$; $^{***}\, q < 0.001$). Accordingly, we first characterize the uncorrected pattern and then verify that the correction leaves the conclusions unchanged.

Figure 2 presents the results as a series of heatmaps, one per network configuration, with agent stubbornness μ_{stub} on the horizontal axis and the opinion prevalence threshold ε_s on the vertical axis. Each cell displays the Cohen's d value of the backfire effect for that parameter combination, with corrected significant cells annotated by asterisks as described above. This layout allows for direct visual comparison of the parameter regions where each network is susceptible to the backfire effect.

The results reveal that the backfire effect is robust across the tested parameter space, remaining largely insensitive to ε_s, increasing with user stubbornness, and emerging only in heavy-tailed networks. In the heavy-tailed networks (CNNR and HK), once stubbornness was at least moderately high ($\mu_{stub} \geq 0.7$), Cohen's d remained positive and predominantly significant across the full ε_s range, indicating that the effect persisted regardless of ε_s. The dependence on μ_{stub} is also the dominant pattern among them. The backfire effect became statistically significant already at moderate stubbornness ($\mu_{stub} \geq 0.6$) in the HK networks, and from $\mu_{stub} \geq 0.7$ in the CNNR networks, with effect sizes broadly increasing up to $\mu_{stub} = 0.9$ (strictly monotonic in CNNR and HK p_t=0.0; HK p_t=0.3 shows a minor non-monotonic dip at $\mu_{stub} = 0.8$), while at the lowest stubbornness ($\mu_{stub} = 0.5$), Cohen's d was near zero or negative

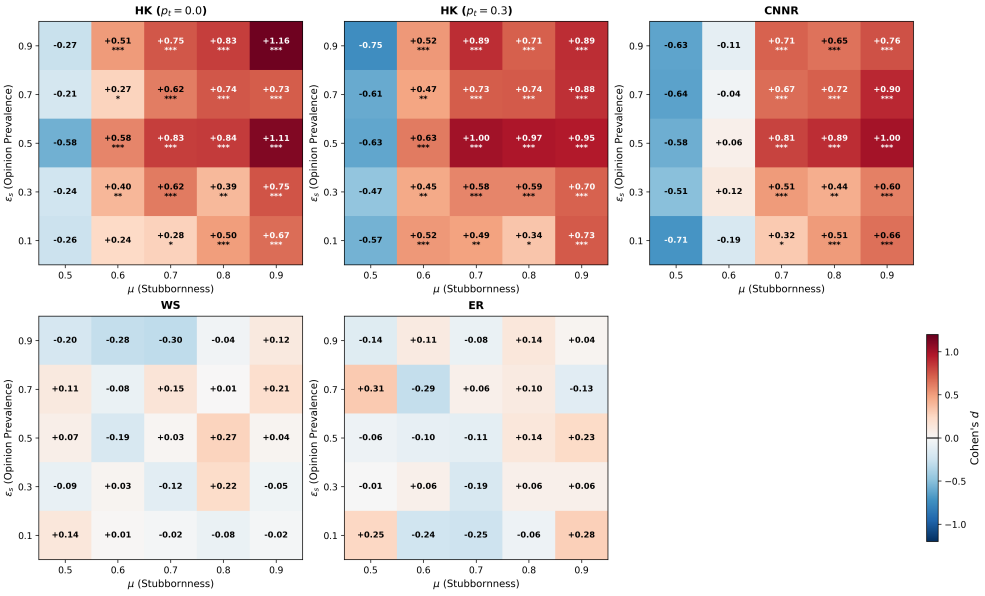


Fig. 2: Sensitivity analysis of the backfire effect across five network configurations (HK with p_t=0.0, HK with p_t=0.3, CNNR, WS, and ER; one heatmap per configuration). Each cell reports the seed-level Cohen's d of the backfire effect b (Eq. (5)) for a combination of agent stubbornness μ_{stub} (horizontal axis) and opinion prevalence threshold ε_s (vertical axis); red (blue) cells indicate a positive (negative) effect. Asterisks mark combinations in which the backfire effect is statistically significant under a one-sided t-test against zero, after Benjamini–Hochberg correction for multiple comparisons applied within each network's 25-cell family (* $q < 0.05$; ** $q < 0.01$; *** $q < 0.001$).

and never significant. By contrast, in ER and WS networks, which lack heavy-tailed degree distributions, the backfire effect was essentially absent. Cohen's d stayed near zero with no coherent dependence on either μ_{stub} or ε_s, and only isolated cells reached significance (3 of 25 in ER and 1 of 25 in WS, at $p < 0.05$).

As shown in Figure 2, correcting for multiple comparisons within each 25-cell family (Benjamini–Hochberg) does not alter this claim. In the heavy-tailed networks, the significant region is mostly unchanged (CNNR: 15/25; HK p_t=0.3: 20/25; HK p_t=0.0: 19/25), with a single HK p_t=0.0 cell at μ_{stub}=0.6 the only one to drop below significance. In ER and WS, the few isolated pre-correction hits all fall to 0/25, so the negative control becomes cleaner rather than weaker. These results suggest that a heavy-tailed (hub-dominated) degree distribution is a critical boundary condition for this backfire phenomenon.

To isolate the manipulated user's own hub status from the aggregate effect of the heavy-tailed topology, we measure a graded dose–response on the continuous target in-degree across the backfire-active regime (the heavy-tailed networks at $\mu_{stub} \geq 0.7$; 45 conditions, 4,294 trials). Binning by within-condition in-degree quartile (Table 5),

Table 5: Dependence of the structural backfire effect on the in-degree of the manipulated hub, pooled across the backfire-active regime (heavy-tailed networks HK with $p_t \in \{0.0, 0.3\}$ and CNNR, at $\mu_{stub} \geq 0.7$; 45 conditions, 4,294 trials).

Target in-degree	n	mean b	Cohen's d	p
All targets (baseline)	4294	+0.059	0.66	$< 10^{-160}$
Q1 (lowest)	1124	+0.018	0.42	$< 10^{-20}$
Q2	1038	+0.026	0.57	$< 10^{-30}$
Q3	1096	+0.046	0.80	$< 10^{-59}$
Q4 (highest)	1036	+0.149	1.24	$< 10^{-100}$

the per-trial b rises monotonically from $\bar{b} = +0.018$ (bottom) to +0.149 (top), and correlates positively with in-degree within every condition (Spearman $\rho > 0$ in 45/45; pooled within-condition $\rho = +0.27$, $n = 4{,}294$, $p < 10^{-70}$) while ER and WS, whose most-connected nodes are not genuine hubs, show no effect at all. All per-quartile tests remain significant under a Benjamini–Hochberg correction across the five reported comparisons (all $q < 10^{-19}$). Backfire is thus governed by the structural prominence of the manipulated hub, acting through the channels documented above (Section 4.4), consistent with the literature identifying centrality as the primary determinant of systemic influence [38, 39].

In summary, the backfire effect is robust against variations in the threshold of opinion prevalence, but is sensitive to agent stubbornness, being most pronounced in heavy-tailed networks (CNNR and HK) when agents maintain a high resistance to opinion change. The within-regime dose–response on target in-degree further confirms that the structural prominence of the manipulated user is the proximate driver of the effect.

5 Discussion

Within the scope of our model, these results identify two boundary conditions for the backfire effect: (1) the heavy-tailed (hub-dominated) nature of the network structure and (2) a high degree of agent stubbornness (or resistance to opinion change). Both conditions are characteristic of many modern social media platforms, which suggests, though does not by itself establish, that such networks may be susceptible to the effect.

We posit that the first condition, a heavy-tailed (hub-dominated) degree distribution, is essential because the backfire mechanism relies on the disproportionate influence of hub users. In social media ecosystems, "influencers" with enormous followings coexist with the vast majority of users who have few connections, forming a heavy-tailed degree distribution. This structural disparity creates fragile connectivity; when a hub acts as a bridge between opposing views, its change in state can trigger a cascade of disconnections, isolating followers into echo chambers, as the causal-chain analysis in Section 4.4 clarifies. The consistent susceptibility of both CNNR and HK

networks, despite their differing clustering properties, further confirms that this hub-dominated degree structure, rather than local clustering, is the decisive structural condition.

Regarding the second condition, we infer that high stubbornness is a prerequisite for the formation and maintenance of echo chambers. As discussed in Section 4.4, the backfire effect is intrinsically linked to the echo chamber formation mechanism. When agents hold their opinions firmly, they are more likely to reject conflicting information from opinion leaders and disconnect from them rather than conform to them. Furthermore, our results indicate that the backfire effect broadly strengthens with stubbornness over $\mu_{stub} \in [0.6, 0.9]$: averaged across the three heavy-tailed networks, mean Cohen's d rises from 0.29 at $\mu_{stub} = 0.6$ to its maximum of 0.83 at $\mu_{stub} = 0.9$, essentially tying at 0.65 and 0.66 for $\mu_{stub} = 0.7$ and 0.8 respectively. This aggregate trend is not strictly monotonic network-by-network: CNNR and HK (p_t=0.0) rise without interruption across the full range, whereas HK (p_t=0.3)—the network used as the canonical mechanistic baseline in Section 4.4—dips slightly at $\mu_{stub} = 0.8$ ($d = 0.74$ at 0.7 versus 0.67 at 0.8) before rising again to 0.83 at 0.9. We therefore describe the relationship as a broadly increasing trend rather than a strict monotonic law. This implies that a network becomes increasingly vulnerable as users hold their intrinsic beliefs more firmly, so long as they retain residual susceptibility to social influence. Our findings suggest that this specific psychological trait, when coupled with the network structure, substantially amplifies the systemic vulnerability to the backfire effect triggered by sudden extreme stance expressions from moderate influencers.

Finally, our results indicate that the threshold for opinion prevalence ε_s, which indirectly represents agent conformity to local opinions via Eq. (2), is not the primary driver of the backfire effect. Intuitively, one might assume that a higher sensitivity to social pressure (high ε_s) would lead to more intense activation, and thus to a stronger backlash. However, as discussed in Section 4.4, the backfire effect that we have discussed in this paper is driven not so much by the magnitude of user activity as by the structural disconnection of ties (unfollowing). Consequently, the phenomenon remains robust, even when agents are not highly sensitive to social pressure.

The temporal dynamics of the backfire effect also merit discussion. As demonstrated in Section 4.3, the post-manipulation trajectory exhibits two distinct phases. Immediately after the onset of manipulation, the targeted camp shows a brief transient uptick in posting activity, the expected short-lived response to the hub's initial opinion shift. However, this transient response is quickly overtaken: within a short window, the opposing camp's log response surpasses that of the targeted camp and *remains* elevated for the remainder of the run. Crucially, therefore, the backfire effect itself is not transient; it is the targeted-camp spike that is short-lived, and what persists is the sustained dominance of the opposing camp. This sustained behavioral signal is further compounded by the underlying structural damage. The network fragmentation induced by the backfire effect—most notably the loss of cross-partisan ties between the hub and its former opposing followers—persists for the remainder of the simulated horizon, well after posting volumes stabilize, and these ties do not reform within our observation window. This indicates that the structural consequences are at least as durable as the already-persistent surface-level behavioral signal.

Several limitations constrain the scope of these results. First, our claims are internal to the model: the backfire effect is established as a robust property of a particular class of opinion dynamics models rather than as a directly observed empirical regularity. Although we anchored the pre-manipulation equilibrium to three stylized facts, this evidence is qualitative and partly directional, draws upon a modest number of empirical studies covering specific platforms and time windows, and does not validate the post-manipulation dynamics themselves, for which comparable ground-truth data remain scarce. This asymmetry is itself informative: whereas the pre-manipulation equilibrium can be anchored to established stylized facts, to the best of our knowledge, comparable empirical regularities have not yet been sufficiently documented for the post-manipulation regime, the very window in which the structural backfire effect unfolds. We therefore treat this gap not merely as a caveat but as a methodological opening for future work: it identifies a concrete need to establish post-intervention stylized facts and to specify observable structural signatures of the mechanism we describe (for instance, hub-loss-driven fragmentation or indices of echo-chamber deepening) that could be measured in longitudinal platform traces, thereby offering a stepping stone toward the empirical validation of post-manipulation dynamics. Second, the model is simplified for tractability. For instance, opinion is abstracted as a single, one-dimensional issue, and the feed is governed solely by follow ties; we deliberately omit algorithmic recommendation and curation, which demonstrably shape exposure on contemporary platforms and could either amplify or dampen the structural cascade we describe. Despite these abstractions, the model reproduces three independent stylized facts of online social media (Section 4.2), suggesting that it captures the essential structure relevant to the mechanism investigated here. Third, the manipulation is reduced to a single, exogenously driven hub whose opinion is displaced deterministically. This design isolates the structural mechanism but does not capture coordinated multi-actor influence operations or the endogenous heterogeneity of real radicalization trajectories.

6 Conclusion

The structural backfire effect refers to the counterintuitive phenomenon whereby, in online social networks, an opinion shift by a moderately opinionated, influential hub toward an extreme stance energizes the opposing camp rather than the targeted one. This study establishes this effect as a robust phenomenon and identifies the underlying mechanism. Building on our preliminary study, and using an agent-based model (ABM) grounded in well-established social-interaction protocols with the manipulation modeled as an incremental opinion shift, we provide direct mechanistic evidence for the causal chain underlying the effect. Furthermore, we delineate its boundary conditions through a sensitivity analysis spanning four network-generating models and a two-parameter grid of agent stubbornness and opinion prevalence thresholds, and establish the model's behavioral credibility against empirical stylized facts drawn from real-world datasets.

The backfire effect proved robust to variations in the threshold of opinion prevalence (the local conforming rate) across its entire tested range, indicating that the

phenomenon is driven by structural reorganization rather than by local conformity pressure. Its emergence instead hinges on two boundary conditions: a heavy-tailed, hub-dominated degree distribution and a relatively high degree of agent adherence to intrinsic opinions. The effect strengthens monotonically with stubbornness and was essentially absent in networks lacking a heavy tail. Restricting the analysis to genuinely high-degree targets sharpens this pattern, demonstrating that the structural prominence of the shifted hub is the proximate driver of the effect. Direct measurement of the causal chain further showed that the effect operates through the deepening of echo chambers via hub loss: the manipulated hub is selectively abandoned by its opposing followers, severing the cross-partisan bridge it provided and collapsing those former followers, but not agents who never relied on it, into homogeneous feeds, thereby deepening their echo chambers.

Addressing the limitations discussed in Section 5 constitutes a natural and important direction for future work, in particular through comparative validation against empirical traces of real-world influence operations, the incorporation of algorithmic curation, and the exploration of richer, multi-actor manipulation scenarios. Notwithstanding these caveats, our findings consistently identify a joint role of hub-dominated structure and user stubbornness within this class of models, offering a hypothesis-generating framework for anticipating when well-intentioned or adversarial interventions on influential users might, in principle, perversely deepen polarization. Insofar as real platforms combine these two features, this candidate mechanism may warrant caution before such interventions are undertaken, and is worth testing against longitudinal platform data ahead of informing the design of more resilient web-communication systems.

Declarations

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Competing interests

The authors have no competing interests to declare that are relevant to the content of this article.

Ethics approval

Not applicable.

Data availability

The simulation data supporting the findings of this study can be reproduced from the publicly available code (see Code availability below). The third-party empirical data

reanalyzed in Section 4.2 are the publicly released account-level data of Buntain and Snegovaya [59].

Code availability

The modified implementations of all algorithms used in the experiments of this study can be found at https://github.com/tooooomoya/structural-backfire-effect.

Author contribution

Conceptualization: TT, FT and TS; Methodology: TT; Implementation and experimental analysis: TT; Writing - original draft preparation: TT; Writing - review and editing: TS and FT; Supervision: TS.

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Appendix A Retained Sample Sizes in the Sensitivity Analysis

For each of the 125 parameter×network conditions in Section 4.5, the realized sample size is $N = \min(n_{\text{valid}}, 50)$, where n_{valid} is the number of seeds (out of the raw pool of 80–100 per network) that produced a qualifying manipulation target (Section 3). Table A1 summarizes N by network. The overall minimum is 37, occurring at HK (p_t=0.3, μ_{stub}=0.8, ε_s=0.5); the canonical baseline used in Section 4.4 (μ_{stub}=0.7) retains 43 seeds. Because each condition's Cohen's d and one-sided t-test are computed solely from its own realized sample and are never compared against another network or condition using a formal cross-condition statistic, this variation does not bias any conclusion drawn in Section 4.5. That ER and WS nonetheless yield qualifying candidates in most conditions reflects the relative nature of the 10%-follower threshold (Section 3) rather than genuine hub structure: these candidates are rarely articulation points (0–7%, versus 52–74% in CNNR/HK), consistent with their lacking a structural bridge to lose.

Table A1: Realized per-condition sample size ($N = \min(n_{\text{valid}}, 50)$) across the sensitivity analysis of Section 4.5.

Network	Conditions	At full N=50	Below 50	Minimum N
CNNR	25	25	0	50
HK (p_t=0.0)	25	25	0	50
HK (p_t=0.3)	25	9	16	37
ER	25	20	5	38
WS	25	24	1	46

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